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Task 2: Case Study

Problem Statement

Choose a real-world case study such as a **Hotel Management System**, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World Case Study

Hotel Management System

A **Hotel Management System (HMS)** is a software application designed to simplify and automate the daily operations of a hotel. It enables guests to book rooms online, check room availability, make secure payments, and manage reservations efficiently. Hotel staff can handle check-in/check-out processes, room allocation, housekeeping, billing, and customer records, while administrators can monitor hotel operations, manage employees, and generate reports for better decision-making.

1. Functional Requirements

Guest

- Register and Login
- Search available rooms
- View room details
- Book hotel rooms
- Cancel or modify reservations
- Make online payments
- View booking history
- Track booking status
- Submit ratings and reviews
- Receive booking confirmations

Receptionist

- Login securely
- Manage guest check-in
- Manage guest check-out
- Allocate rooms
- Update room availability

- Generate bills
- Handle customer inquiries

Admin

- Manage hotel rooms
- Manage room categories
- Update room availability
- Manage customer records
- Manage staff information
- Generate occupancy reports
- Generate revenue reports
- Manage discounts and offers

Housekeeping Staff

- Login securely
- View assigned rooms
- Update room cleaning status
- Report maintenance issues

2. User Stories

User ID	User Story	Priority
US-01	As a guest, I want to register so that I can book hotel rooms online.	High
US-02	As a guest, I want to log in securely to access my bookings.	High
US-03	As a guest, I want to search rooms based on availability and room type.	High
US-04	As a guest, I want to book a hotel room online.	High
US-05	As a guest, I want to modify or cancel my reservation.	Medium
US-06	As a guest, I want to make secure online payments.	High
US-07	As a guest, I want to view my booking history.	Medium
US-08	As a guest, I want to rate my hotel stay.	Medium
US-09	As an admin, I want to manage hotel rooms and room categories.	High
US-10	As an admin, I want to manage customer records.	High

User ID	User Story	Priority
US-11	As an admin, I want to generate occupancy and revenue reports.	Medium
US-12	As a receptionist, I want to manage guest check-in and check-out.	High
US-13	As a housekeeping staff member, I want to update room cleaning status.	High
US-14	As a housekeeping staff member, I want to report maintenance issues.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Hotel requirements may change based on customer feedback.
- New services and room types can be added continuously.
- Faster implementation of new features.
- Continuous testing and bug fixing.
- Regular customer and staff feedback.
- Suitable for both web and mobile hotel management applications.

Sprint Plan

Sprint	Features
Sprint 1	User Registration & Login
Sprint 2	Room Management & Search
Sprint 3	Room Booking & Reservation
Sprint 4	Check-in, Check-out & Billing
Sprint 5	Online Payment & Notifications
Sprint 6	Admin Dashboard & Report Management
Sprint 7	Housekeeping & Maintenance Module
Sprint 8	Testing & Deployment

4. Programming Language Selection

Backend

Java (Spring Boot)

Reasons

- Secure
- Scalable
- High Performance
- REST API Development
- Easy Database Integration
- Suitable for Enterprise Applications

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

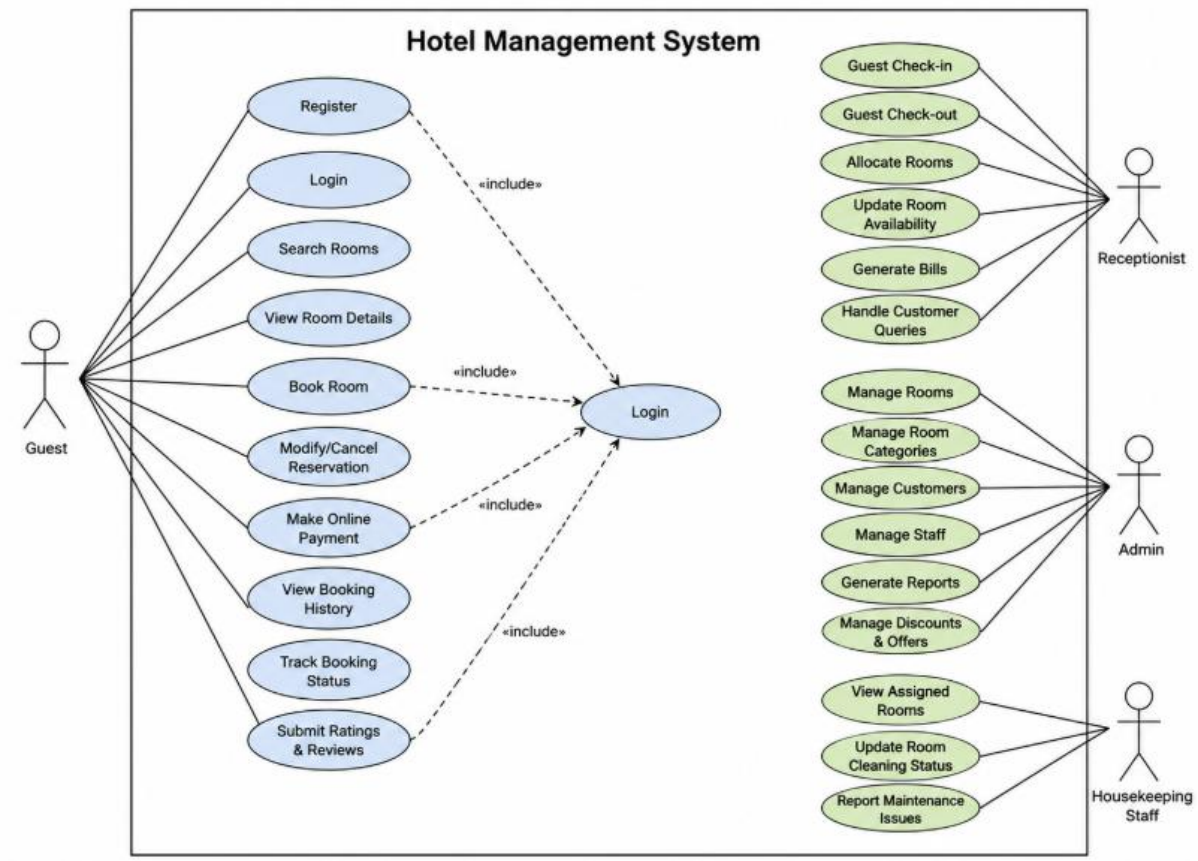
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Razorpay / Stripe Payment Gateway
- Email Notification Service
- Git & GitHub
- Docker
- AWS Cloud

5. System Architecture



6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather guest, receptionist, admin, and housekeeping requirements.
- Create Software Requirement Specification (SRS).
- Identify functional and non-functional requirements.
- Prepare Use Case Diagrams.

Phase 2 – System Design

- Design database schema.
- Design user interface (UI) wireframes.
- Create UML diagrams.

- Develop system architecture.
- Design room booking and billing workflow.

Phase 3 – Development

- Implement user registration and authentication.
- Develop room search and booking module.
- Create reservation management.
- Implement check-in and check-out process.
- Integrate online payment gateway.
- Develop billing system.
- Build admin dashboard.
- Develop housekeeping management module.

Phase 4 – Testing

Unit Testing

- User Registration
- Login
- Room Search
- Room Booking
- Online Payment

Integration Testing

- Booking + Payment
- Reservation + Room Availability
- Billing + Check-out
- Housekeeping + Room Status

System Testing

- Complete Hotel Management System Testing
- Performance Testing
- Compatibility Testing

User Acceptance Testing (UAT)

- Verify guest requirements.

- Verify hotel staff workflows.
- Collect user feedback before deployment.

Phase 5 – Deployment

- Docker Container
- AWS EC2
- MySQL Database
- Nginx Web Server
- Cloud Deployment and Monitoring

7. Evaluation Strategy

Evaluation Parameter	**Method**
Functional Accuracy	Requirement verification
Performance	Load testing using JMeter
Security	Authentication and penetration testing
Reliability	Continuous monitoring
Usability	User feedback survey
Scalability	Concurrent user testing
Maintainability	Code review and documentation
Room Availability Accuracy	Verify room status after every booking
Booking Efficiency	Measure reservation processing time
Billing Accuracy	Verify invoice generation and payment records

8. Expected Outcomes

- Secure hotel room booking platform.
- Fast room search and reservation process.
- Reliable room allocation and booking management.
- Accurate room availability updates.
- Secure online payment system.

- Efficient check-in and check-out management.
- Improved housekeeping coordination.
- Better customer satisfaction through smooth hotel services.
- Scalable architecture for future hotel expansion.
- Easy management of rooms, bookings, staff, and reports by the administrator.